Section 8: Docker + Deployment Basics – 10 Marks

Tasks:

1. Create a **Dockerfile** for your API
2. Run container locally
3. Explain:
    - Environment variables
    - Virtual environment vs Docker

Answer:

1. Dockerfile for API

👉 Docker is used to package your app into a container.

```
# Base image
FROM python:3.10

# Set working directory
WORKDIR /app

# Copy files
COPY . .

# Install dependencies
RUN pip install flask

# Expose port
EXPOSE 5000

# Run app
CMD ["python", "app.py"]
```

2. Run Container Locally

Step 1: Build image

```
docker build -t myapp .
```

Step 2: Run container
docker run -p 5000:5000 myapp

Now open browser: http://localhost:5000

3. Explanation

🔸 **Environment Variables**

- Used to store configuration like:
  - API keys
  - Database URL
- Keeps sensitive data outside code

Virtual Environment vs Docker

Feature	Virtual Environment	Docker
Scope	Python only	Full system
Isolation	Libraries	OS + Libraries
Setup	venv	Dockerfile
Use case	Development	Deployment

👉 *Docker is used to containerize applications, allowing them to run consistently across environments, unlike virtual environments which only manage Python dependencies.*